

Module 4: Chromosomes & Cell Division

Comprehensive Identification & Multiple-Choice Reviewer — 40 Items

Name: _____

Date: _____ Score: _____

INSTRUCTIONS: Read each statement carefully. Identify the correct term or concept described in the blank (____). Choose the letter of the correct answer from the choices provided (a, b, c, or d) and write it on the blank provided. An exhaustive Answer Key with detailed rationales is provided at the end of this document.

PART I: IDENTIFICATION QUESTIONNAIRE (40 ITEMS)

- _____ 1. The condensed form of DNA in a cell that surrounds and wraps around spool-like proteins called histones is known as what structure?
a.) Centriole b.) Ribosome c.) Chromosome d.) Nucleolus
- _____ 2. What proteins serve as the molecular spools around which DNA coils and condenses into chromatin and chromosomes?
a.) Spindle fibers b.) Histones c.) Centrosomes d.) Ribosomes
- _____ 3. A specific sequence of DNA that codes for a protein with a structural or functional role in the cell is defined as a what?
a.) Codon b.) Centromere c.) Gene d.) Histone
- _____ 4. How many different kinds of chromosomes are present in human cells?
a.) 23 kinds b.) 22 kinds c.) 46 kinds d.) 2 kinds
- _____ 5. According to findings from the Human Genome Project, approximately how many base pairs are in the human genome?
a.) 23 million base pairs b.) 3.2 million base pairs c.) 3.2 billion base pairs d.) 46 billion base pairs
- _____ 6. What percentage of our modern human DNA is unique to modern humans and distinct from early human ancestors?
a.) 1% b.) 23% c.) 7% d.) 99%
- _____ 7. All cells within the human body except for reproductive cells (eggs and sperm) are classified as what type of cells?
a.) Zygotes b.) Gametes c.) Sex cells d.) Somatic cells
- _____ 8. What is the total diploid chromosome number present in normal human somatic cells?
a.) 46 chromosomes b.) 23 chromosomes c.) 92 chromosomes d.) 44 chromosomes
- _____ 9. What term describes human reproductive cells (eggs and sperm) that contain only one set of 23 chromosomes?
a.) Polyploid b.) Haploid c.) Diploid d.) Tetraploid
- _____ 10. Chromosomes that occur in pairs and contain genes that code for or affect the same biological traits are called what?
a.) Heterozygous chromosomes b.) Analogous chromosomes c.) Homologous chromosomes d.) Sister chromatids
- _____ 11. How many homologous chromosome pairs are found in a normal human somatic cell?
a.) 46 pairs b.) 22 pairs c.) 23 pairs d.) 1 pair
- _____ 12. The single pair of human chromosomes that determines biological gender is designated as what?
a.) Histones b.) Autosomes c.) Centrosomes d.) Sex chromosomes
- _____ 13. Which sex chromosome combination corresponds to biological males in humans?

a.) XO b.) XY c.) XX d.) YY

_____ 14. Which sex chromosome combination corresponds to biological females in humans?

a.) YY b.) XY c.) XX d.) XXY

_____ 15. The 22 pairs of human chromosomes that do not determine gender are called what?

a.) Chromatids b.) Autosomes c.) Centrosomes d.) Sex chromosomes

_____ 16. Which type of nuclear division produces two genetically identical daughter cells with the same number of chromosomes as the parent cell for growth and tissue repair?

a.) Binary fission b.) Mitosis c.) Meiosis d.) Fertilization

_____ 17. Which type of nuclear division reduces the chromosome number by half and occurs specifically during the production of gametes?

a.) Cytokinesis b.) Interphase c.) Mitosis d.) Meiosis

_____ 18. What are the two major phases that comprise the eukaryotic cell cycle?

a.) Prophase and telophase b.) Mitosis and meiosis c.) Interphase and cell division d.) G1 and G2

_____ 19. The period between cell divisions during which the cell grows and replicates its DNA and organelles is called what?

a.) Anaphase b.) Cytokinesis c.) Interphase d.) Metaphase

_____ 20. What are the three sequential sub-phases of interphase?

a.) G1, S, and G2 b.) G0, G1, and G2 c.) Mitosis, meiosis, and cytokinesis d.) Prophase, metaphase, and anaphase

_____ 21. During which sub-phase of interphase does major cell growth occur before DNA synthesis begins?

a.) S phase b.) G1 phase c.) G2 phase d.) M phase

_____ 22. During which specific sub-phase of interphase does DNA synthesis (replication) take place?

a.) G2 phase b.) Prophase c.) G1 phase d.) S phase

_____ 23. During which sub-phase of interphase does the cell undergo a second stage of growth in final preparation for mitosis?

a.) G2 phase b.) Cytokinesis c.) G1 phase d.) S phase

_____ 24. In which phase of mitosis does chromatin condense into visible chromosomes, the nuclear membrane break down, and centrioles move to opposite poles?

a.) Telophase b.) Anaphase c.) Prophase d.) Metaphase

_____ 25. Why is the condensation of DNA around histones during prophase critical for cell division?

a.) It makes it easier to separate sister chromatids without breaking them b.) It dissolves the cell membrane c.) It synthesizes new spindle fibers d.) It speeds up DNA replication

_____ 26. During which phase of mitosis do chromosomes attach to spindle fibers and line up along the equator (center) of the mitotic spindle?

a.) Prophase b.) Metaphase c.) Telophase d.) Anaphase

_____ 27. During which phase of mitosis do sister chromatids split at the centromere and get pulled toward opposite poles of the cell?

a.) Metaphase b.) Telophase c.) Prophase d.) Anaphase

_____ 28. What specialized chromosomal region holds sister chromatids together until it splits during anaphase?

a.) Centrosome b.) Telomere c.) Centriole d.) Centromere

_____ 29. In which phase of mitosis does a nuclear envelope reform around each group of chromosomes, the spindle disassemble, and chromosomes decondense?

a.) Anaphase b.) Telophase c.) Prophase d.) Metaphase

_____ 30. What is the separate cellular process involving the physical division of the cytoplasm into two daughter cells?

a.) Cytokinesis b.) Karyokinesis c.) Interphase d.) Apoptosis

_____ 31. At what stage of mitosis does cytokinesis usually begin in human cells?

a.) Metaphase b.) Prophase c.) Telophase d.) G1 phase

- _____ **32.** What cytoskeletal structure contracts at the midline of a dividing animal cell to create a cleavage furrow during cytokinesis?
a.) Microtubules b.) Histone octamer c.) Centrioles d.) A band of microfilaments
- _____ **33.** What groove or indentation forms at the cell equator as contracting microfilaments deepen to pinch the cell into two?
a.) Cleavage furrow b.) Centromere c.) Equatorial plate d.) Cell plate
- _____ **34.** Skin, bone, blood, and internal organ cells are all examples of what category of cells?
a.) Somatic cells b.) Gametes c.) Germ cells d.) Sex cells
- _____ **35.** How many autosomes (individual non-sex chromosomes) are present in a human somatic cell?
a.) 46 autosomes b.) 44 autosomes c.) 22 autosomes d.) 23 autosomes
- _____ **36.** Which cellular organelle pairs migrate away from each other toward opposite poles during prophase to organize the spindle apparatus?
a.) Golgi bodies b.) Lysosomes c.) Ribosomes d.) Centrioles
- _____ **37.** What structure is composed of microtubules that attach to chromosomes and direct their alignment and separation during mitosis?
a.) Cell cortex b.) Mitotic spindle c.) Cleavage furrow d.) Nuclear envelope
- _____ **38.** What happens to the mitotic spindle during telophase?
a.) It disassembles and disappears b.) It pulls chromatids to the center c.) It duplicates d.) It forms the cell membrane
- _____ **39.** In which stage of the cell cycle does a cell spend the majority of its functional life performing daily metabolic activities and preparing for division?
a.) Interphase b.) Cytokinesis c.) Mitosis d.) Anaphase
- _____ **40.** What is the primary biological outcome of mitosis followed by cytokinesis?
a.) A single tetraploid cell b.) Four genetically diverse haploid gametes c.) Two non-identical haploid cells d.) Two genetically identical diploid daughter cells

COMPLETE ANSWER KEY & EXPLANATIONS

Module 4: Chromosomes & Cell Division — Reference Key with Concept Explanations

Item	Key	Correct Answer	Curriculum Explanation & Reference
1	C	Chromosome	[Forms of Chromosomes] A chromosome contains DNA in its condensed form wrapped around proteins called histones.
2	B	Histones	[Forms of Chromosomes] DNA wraps around proteins called histones to form compact chromatin and chromosomes, preventing breakage during cell division.
3	C	Gene	[Forms of Chromosomes] A gene is a sequence of DNA arranged in a specific manner that codes for a protein with a structural or functional role in the cell.
4	A	23 kinds	[Forms of Chromosomes] Each of the 23 different kinds of chromosomes in human cells contains a specific sequence of genes.
5	C	3.2 billion base pairs	[Human Genome Project] Researchers have sequenced approximately 3.2 billion base pairs in the human genome (about 99% of the genome).
6	C	7%	[Human Genome Project] Out of the 99% of sequenced human genome, 7% of our DNA is unique to modern humans compared to early ancestors.
7	D	Somatic cells	[Somatic Cells] Somatic cells refer to all cells within the human body and are diploid (46 chromosomes) except eggs and sperm.
8	A	46 chromosomes	[Somatic Cells] Somatic cells are diploid, containing 46 chromosomes (23 homologous pairs).
9	B	Haploid	[Gametes] Eggs and sperm are haploid, meaning they contain only one set of 23 chromosomes.
10	C	Homologous chromosomes	[Homologous Chromosomes] Homologous chromosomes are pairs that contain genes coding for or affecting the same traits, one from each parent.
11	C	23 pairs	[Homologous Chromosomes] The diploid number of 46 chromosomes translates to 23 homologous pairs in each somatic cell.
12	D	Sex chromosomes	[Sex Chromosomes] One pair of chromosomes determines gender and is called the sex chromosomes (XY in males, XX in females).
13	B	XY	[Sex Chromosomes] Males have XY chromosomes, while females have XX chromosomes.
14	C	XX	[Sex Chromosomes] Females possess two X chromosomes (XX).
15	B	Autosomes	[Autosomes] There are 22 pairs of non-sex chromosomes, which are

termed autosomes.

16	B	Mitosis	[Types of Cell Division] Mitosis creates daughter cells with the exact same number of chromosomes as the original cell, functioning in growth and tissue repair.
17	D	Meiosis	[Types of Cell Division] Meiosis creates cells with half the number of chromosomes (haploid) as the original cell and occurs during gamete production.
18	C	Interphase and cell division	[Cell Cycle] The cell cycle consists of two major phases: interphase and cell division (mitosis + cytokinesis).
19	C	Interphase	[Cell Cycle] Interphase is the period between cell divisions during which DNA and organelles replicate in preparation for division.
20	A	G1, S, and G2	[Interphase] Interphase is divided into G1 (First Gap), S (Synthesis), and G2 (Second Gap) phases.
21	B	G1 phase	[Interphase] G1 (First Gap) is the stage where major cell growth occurs before DNA synthesis begins.
22	D	S phase	[Interphase] During the S phase (Synthesis), DNA replication occurs, producing identical copies of genetic material.
23	A	G2 phase	[Interphase] In the G2 phase, another stage of growth and protein synthesis occurs before mitosis begins.
24	C	Prophase	[Mitosis Phases] During prophase, chromatin condenses, the nuclear membrane breaks down, and centrioles migrate toward opposite ends of the cell.
25	A	It makes it easier to separate sister chromatids without breaking them	[Mitosis Phases] DNA wrapping tightly around histones prevents long DNA strands from tangling and breaking during chromatid separation.
26	B	Metaphase	[Mitosis Phases] During metaphase, chromosomes attach to spindle fibers and migrate to the equator (metaphase plate) of the mitotic spindle.
27	D	Anaphase	[Mitosis Phases] During anaphase, sister chromatids separate at the centromere and are pulled toward opposite cell poles.
28	D	Centromere	[Mitosis Phases] Chromosomes split at the centromere during anaphase as spindle fibers pull them apart.
29	B	Telophase	[Mitosis Phases] During telophase, nuclear envelopes reform, the mitotic spindle disassembles, and chromosomes decondense back into chromatin.
30	A	Cytokinesis	[Cytokinesis] Cytokinesis is the process involving the division of cytoplasm, beginning during telophase.

31	C	Telophase	[Cytokinesis] Cytokinesis usually begins sometime during telophase.
32	D	A band of microfilaments	[Cytokinesis] A band of microfilaments located at the cell midline contracts, forming a furrow that pinches the cell in two.
33	A	Cleavage furrow	[Cytokinesis] The contracting microfilament band forms a furrow that pinches the cell into two daughter cells once it deepens.
34	A	Somatic cells	[Somatic Cells] Examples of somatic cells include cells of internal organs, skin, bones, blood, and connective tissues.
35	B	44 autosomes	[Autosomes] There are 22 pairs of autosomes, which equals 44 individual autosomes (plus 2 sex chromosomes = 46 total).
36	D	Centrioles	[Mitosis Phases] During prophase, centrioles move away from each other toward opposite ends of the cell to form the mitotic spindle.
37	B	Mitotic spindle	[Mitosis Phases] The mitotic spindle is composed of spindle fibers that attach to chromosomes at metaphase and pull them apart at anaphase.
38	A	It disassembles and disappears	[Mitosis Phases] During telophase, the mitotic spindle that caused chromosomes to migrate disassembles and disappears.
39	A	Interphase	[Cell Cycle] Interphase is the longest phase of the cell cycle, consisting of G1, S, and G2 phases.
40	D	Two genetically identical diploid daughter cells	[Types of Cell Division] Mitosis distributes replicated genetic material equally between two daughter cells, producing identical diploid cells.